PRINCIPLES OF MACROECONOMICS

Chapter 16: Inflation, Disinflation, and Deflation

Money, Inflation, and Unemployment in the Short Run and Long Run

ECON_OX 112 · November 12, 2026

Guilherme Paiva Pinto

Learning Objectives

- A · Explain seigniorage, the inflation tax, and how people respond to high inflation.
- B · Explain the short-run inflation–unemployment trade-off and how supply shocks change it.
- C · Explain why expectations adjustment leads to a vertical long-run Phillips curve at NAIRU.

16.1 Money and Inflation

- Explain seigniorage, the inflation tax, and how people respond to high inflation.

The Classical Model of Money and Prices

Long-run monetary neutrality

Other conditions fixed, increasing the money supply raises the price level rather than real output.

ECB, Scope of monetary policy

The Classical Model of Money and Prices

Long-run monetary neutrality

Other conditions fixed, increasing the money supply raises the price level rather than real output.

High inflation

Frequent wage and price adjustments make the classical flexible-price approximation useful.

ECB, Scope of monetary policy

The Classical Model of Money and Prices

Long-run monetary neutrality

Other conditions fixed, increasing the money supply raises the price level rather than real output.

High inflation

Frequent wage and price adjustments make the classical flexible-price approximation useful.

Low, stable inflation

With sticky wages and prices, Keynesian models help explain short-run changes in output and employment.

ECB, Scope of monetary policy

Seigniorage and the Inflation Tax

M: state-issued money stock · ΔM: new money issued · P: price level

Seigniorage

Nominal revenue: $\Delta M = M_t - M_{t-1}$. Real revenue: $\Delta M / P_t$.

IMF, Suriname: Toward Stability and Growth (2009), Box 5.2

Seigniorage and the Inflation Tax

M: state-issued money stock · ΔM: new money issued · P: price level

Seigniorage

Nominal revenue: $\Delta M = M_t - M_{t-1}$. Real revenue: $\Delta M / P_t$.

The inflation tax

Rising prices reduce the purchasing power of money held by the public.

IMF, Suriname: Toward Stability and Growth (2009), Box 5.2

Seigniorage and the Inflation Tax

M: state-issued money stock · ΔM: new money issued · P: price level

Seigniorage

Nominal revenue: $\Delta M = M_t - M_{t-1}$. Real revenue: $\Delta M / P_t$.

The inflation tax

Rising prices reduce the purchasing power of money held by the public.

Government finance

Money issuance can finance spending, while the resulting inflation imposes a cost on money holders.

IMF, Suriname: Toward Stability and Growth (2009), Box 5.2

The Logic of Hyperinflation

Money-financed deficits

The government relies on money creation to cover a persistent financing gap.

Nachega, IMF Working Paper 05/221 (2005)

The Logic of Hyperinflation

Money-financed deficits

The government relies on money creation to cover a persistent financing gap.

Avoiding the inflation tax

As money loses purchasing power, people hold fewer real domestic money balances.

Nachega, IMF Working Paper 05/221 (2005)

The Logic of Hyperinflation

Money-financed deficits

The government relies on money creation to cover a persistent financing gap.

Avoiding the inflation tax

As money loses purchasing power, people hold fewer real domestic money balances.

A reinforcing process

If the government issues money faster to obtain the same resources, inflation and avoidance can reinforce each other.

Nachega, IMF Working Paper 05/221 (2005)

Money, Prices, and Wages During Hyperinflation

Function	During hyperinflation
Store of value	People may save in dollars, gold, or goods to limit losses on domestic money.

IMF, Money: At the Center of Transactions · Bundesbank (2012); Bundestag (2023), Germany’s hyperinflation · Bundestag, Hyperinflation in Deutschland 1923

Money, Prices, and Wages During Hyperinflation

Function	During hyperinflation
Store of value	People may save in dollars, gold, or goods to limit losses on domestic money.
Unit of account	Prices and contracts may be quoted in a more stable currency or indexed.

IMF, Money: At the Center of Transactions · Bundesbank (2012); Bundestag (2023), Germany's hyperinflation · Bundestag, Hyperinflation in Deutschland 1923

Money, Prices, and Wages During Hyperinflation

Function	During hyperinflation
Store of value	People may save in dollars, gold, or goods to limit losses on domestic money.
Unit of account	Prices and contracts may be quoted in a more stable currency or indexed.
Medium of exchange	Sellers may prefer foreign currency or another means of payment.

Prices may change daily or within the day; workers seek frequent wage corrections.

Pay intervals may shorten: monthly → biweekly → weekly → daily → within-day.

IMF, Money: At the Center of Transactions · Bundesbank (2012); Bundestag (2023), Germany's hyperinflation · Bundestag, Hyperinflation in Deutschland 1923

16.2 Moderate Inflation and Disinflation

- Explain the short-run inflation–unemployment trade-off and how supply shocks change it.

Moderate Inflation and Disinflation

The political business cycle

Elected leaders may seek expansion before elections to raise output and reduce unemployment.

Keller and May (1984), The Presidential Political Business Cycle of 1972 · Federal Reserve Board, Waller (2023), The Unstable Phillips Curve

Moderate Inflation and Disinflation

The political business cycle

Elected leaders may seek expansion before elections to raise output and reduce unemployment.

The cost of disinflation

Contractionary monetary or fiscal policy reduces demand and can temporarily raise unemployment.

Keller and May (1984), The Presidential Political Business Cycle of 1972 · Federal Reserve Board, Waller (2023), The Unstable Phillips Curve

Moderate Inflation and Disinflation

The political business cycle

Elected leaders may seek expansion before elections to raise output and reduce unemployment.

The cost of disinflation

Contractionary monetary or fiscal policy reduces demand and can temporarily raise unemployment.

The political dilemma

Tolerating inflation may postpone the unpopular loss of output and jobs associated with reducing it.

Keller and May (1984), The Presidential Political Business Cycle of 1972 · Federal Reserve Board, Waller (2023), The Unstable Phillips Curve

Potential Output and Unemployment

Potential output, Y^P

The real GDP the economy can sustain with resources used at normal rates.

Potential Output and Unemployment

Potential output, Y^P

The real GDP the economy can sustain with resources used at normal rates.

The output gap

100 × (actual real GDP − potential real GDP) / potential real GDP

Potential Output and Unemployment

Potential output, Y^P

The real GDP the economy can sustain with resources used at normal rates.

The output gap

100 × (actual real GDP − potential real GDP) / potential real GDP

The unemployment rate

Actual unemployment = natural unemployment + cyclical unemployment.

Okun’s Law

Output gaps and unemployment deviations from the natural rate move in opposite directions.

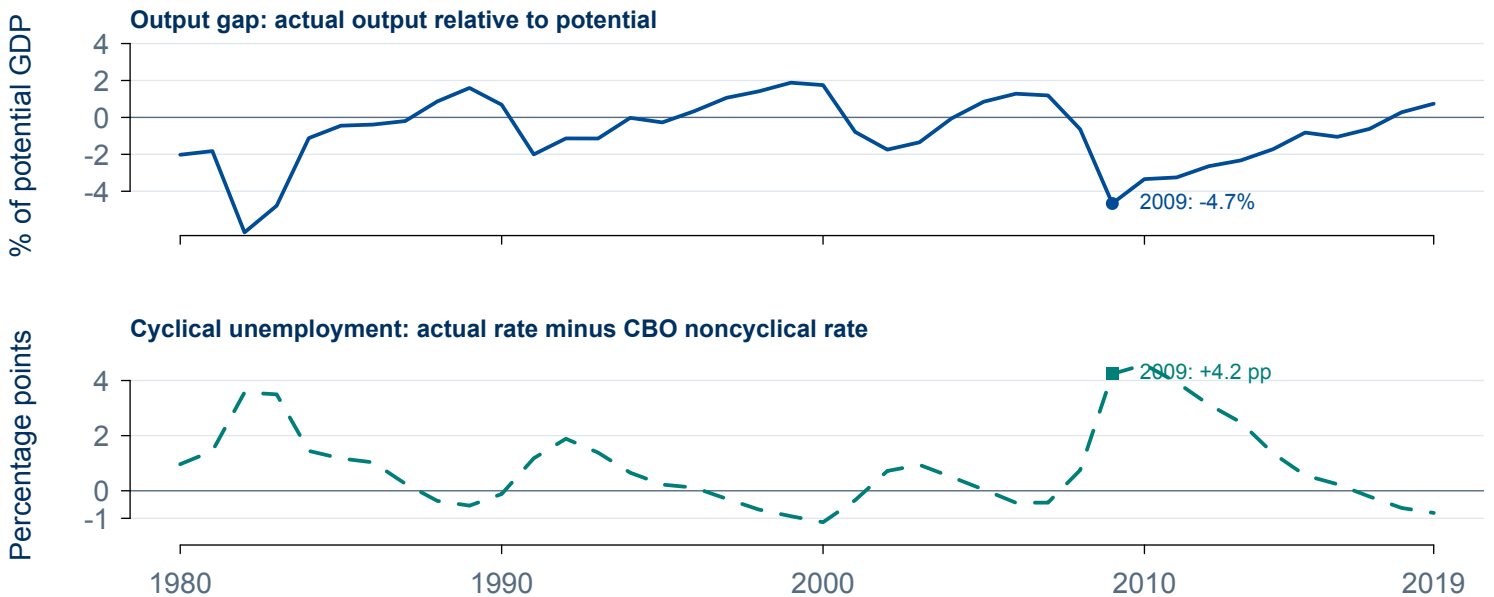
Output	Output gap	Unemployment	Cyclical unemployment
Actual = potential	Zero	Actual = natural	Zero
Actual > potential	Positive	Actual < natural	Negative
Actual < potential	Negative	Actual > natural	Positive

Output gap (%) ≈ −2 × (u − u*)

u*: natural unemployment rate. u − u*: cyclical unemployment, in percentage points. The coefficient is an approximate US rule of thumb.

Federal Reserve Board, Principles for Monetary Policy · Daly and Hobijn, FRBSF (2010), Okun’s Law

Cyclical Unemployment and the Output Gap



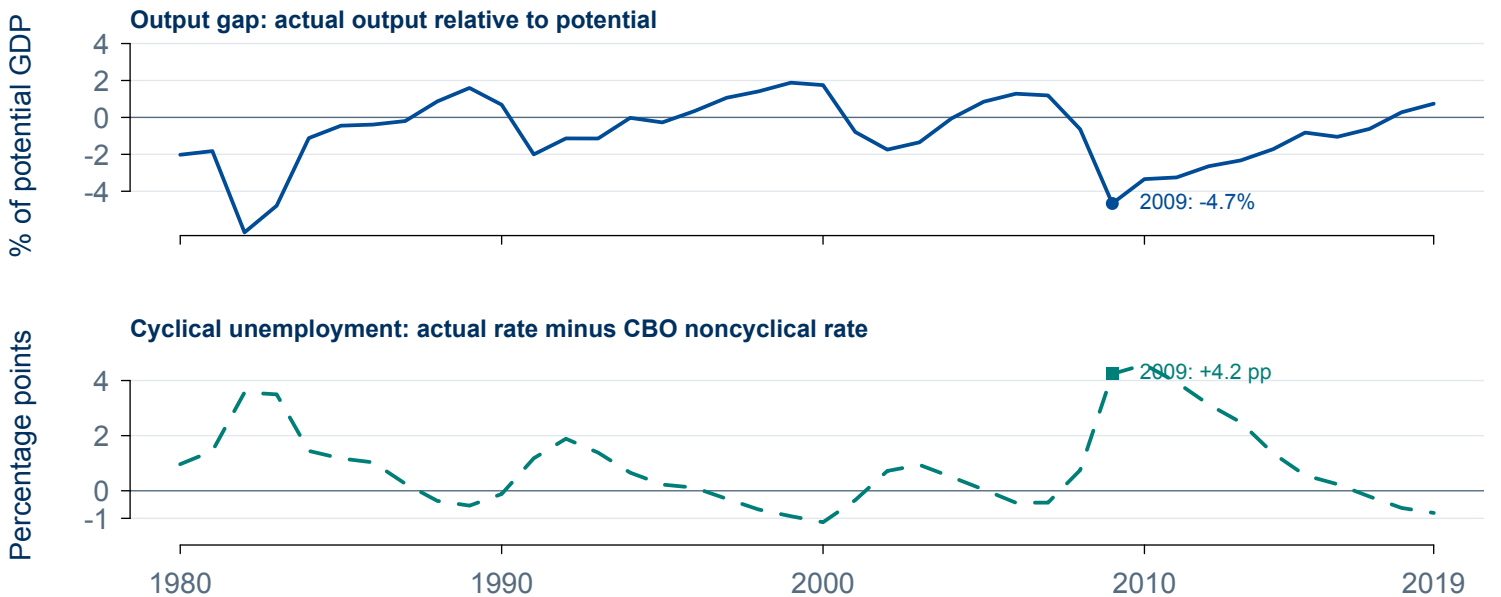
Source: CBO, February 2026 historical economic data; annual values, 1980–2019. Original calculations.

Estimated potential output

CBO combines sustainable labor input, capital services, and productivity.

CBO, February 2026 historical economic data · CBO, Approaches to Estimating the Noncyclical Rate (2023) · CBO, Forecasting methods (2018)

Cyclical Unemployment and the Output Gap



Source: CBO, February 2026 historical economic data; annual values, 1980–2019. Original calculations.

Estimated potential output

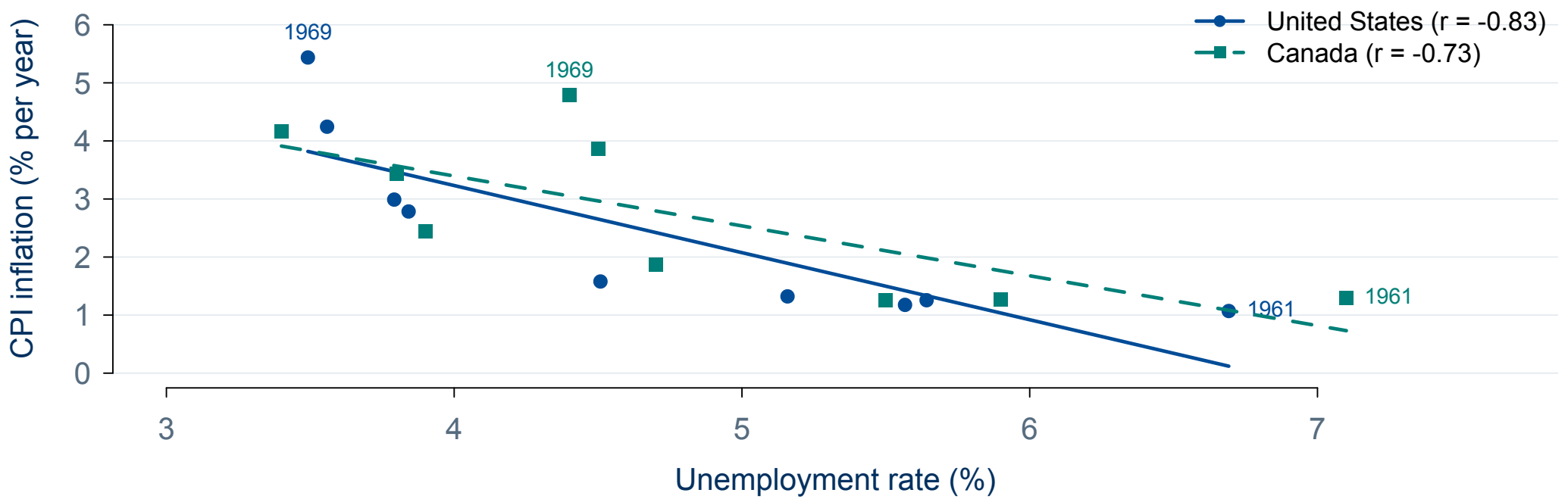
CBO combines sustainable labor input, capital services, and productivity.

Estimated noncyclical unemployment

CBO’s 2023 approach reweights benchmark demographic unemployment rates as the workforce changes.

CBO, February 2026 historical economic data · CBO, Approaches to Estimating the Noncyclical Rate (2023) · CBO, Forecasting methods (2018)

Inflation and Unemployment, 1961–1969



All annual observations, 1961–1969; separate descriptive fits. National definitions differ.

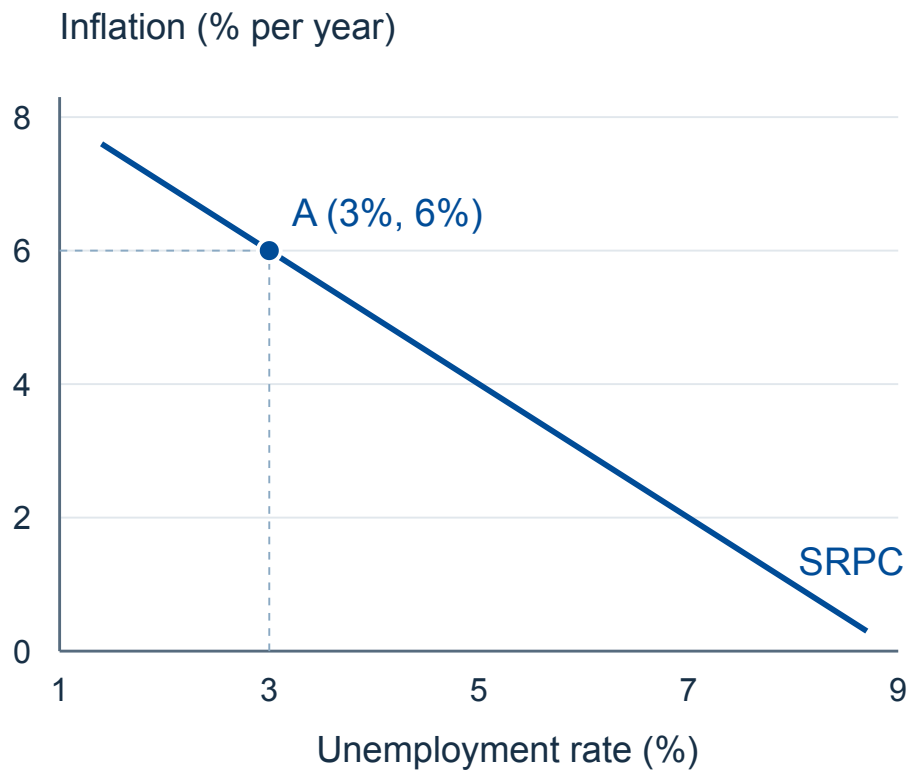
US: CBO/BLS. Adapted from Statistics Canada, CPI Table 18-10-0005-01 and LFS, 1961–1969.

CPI released January 19, 2026; LFS chart February 10, 2017. Statistics Canada Open Licence.

This does not constitute an endorsement by Statistics Canada of this product.

CBO, February 2026 historical economic data · Statistics Canada, CPI 18-10-0005-01 · Statistics Canada Open Licence

The Short-Run Phillips Curve



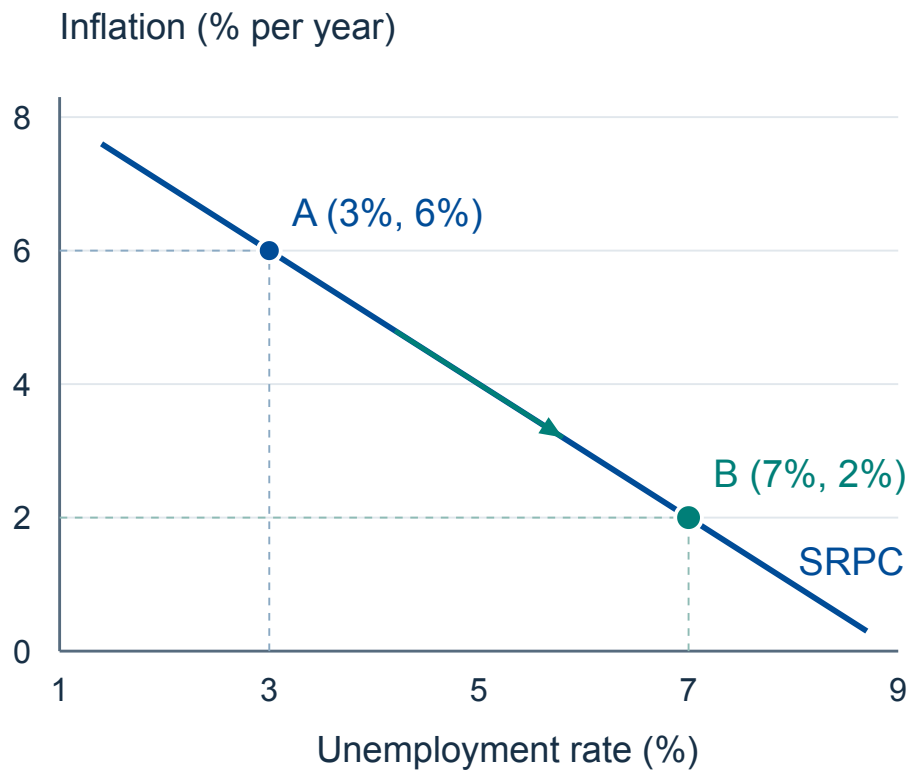
A.W.H. Phillips · New Zealand economist · 1958

The relationship

With expectations and supply conditions fixed, lower unemployment is associated with higher price inflation.

Reserve Bank of New Zealand, Monetary policy objectives (2019) · Federal Reserve Board, Waller (2023), The Unstable Phillips Curve

The Short-Run Phillips Curve



A.W.H. Phillips · New Zealand economist · 1958

The relationship

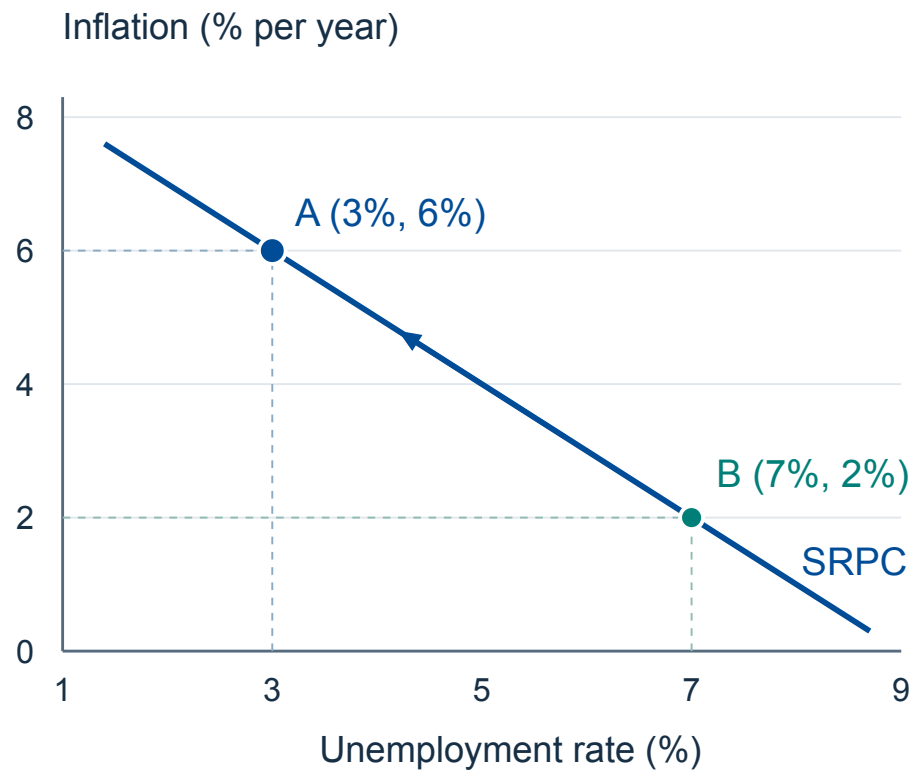
With expectations and supply conditions fixed, lower unemployment is associated with higher price inflation.

A → B: demand restraint

Weaker labor demand limits wage increases. Inflation falls, while unemployment rises.

Reserve Bank of New Zealand, Monetary policy objectives (2019) · Federal Reserve Board, Waller (2023), The Unstable Phillips Curve

The Short-Run Phillips Curve



A.W.H. Phillips · New Zealand economist · 1958

The relationship

With expectations and supply conditions fixed, lower unemployment is associated with higher price inflation.

A → B: demand restraint

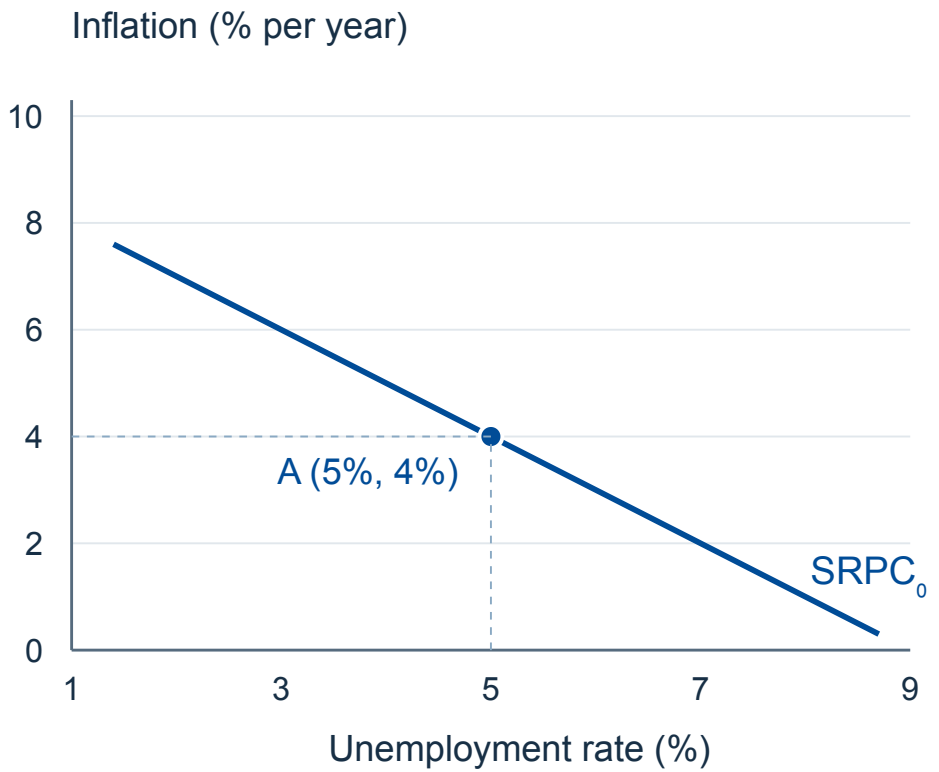
Weaker labor demand limits wage increases. Inflation falls, while unemployment rises.

B → A: demand expansion

A tighter labor market strengthens wage bargaining and adds price pressure; unemployment falls.

Reserve Bank of New Zealand, Monetary policy objectives (2019) · Federal Reserve Board, Waller (2023), The Unstable Phillips Curve

Supply Shocks and the Phillips Curve



A: before an oil shock

Unemployment is 5% and inflation is 4% on $SRPC_0$.

Federal Reserve Board, Waller (2023), The Unstable Phillips Curve · RBA, Causes of Inflation

Supply Shocks and the Phillips Curve



A: before an oil shock

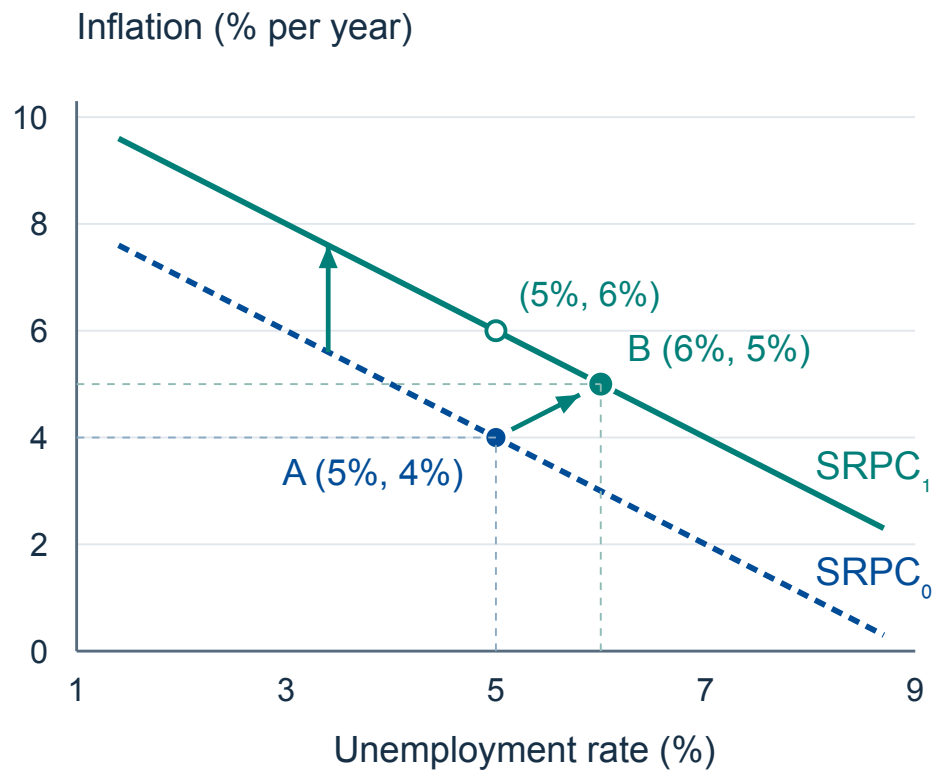
Unemployment is 5% and inflation is 4% on $SRPC_0$.

Higher oil costs: an upward shift

Oil costs raise inflation at each unemployment rate. Falling costs reverse the shift.

Federal Reserve Board, Waller (2023), The Unstable Phillips Curve · RBA, Causes of Inflation

Supply Shocks and the Phillips Curve



A: before an oil shock

Unemployment is 5% and inflation is 4% on $SRPC_0$.

Higher oil costs: an upward shift

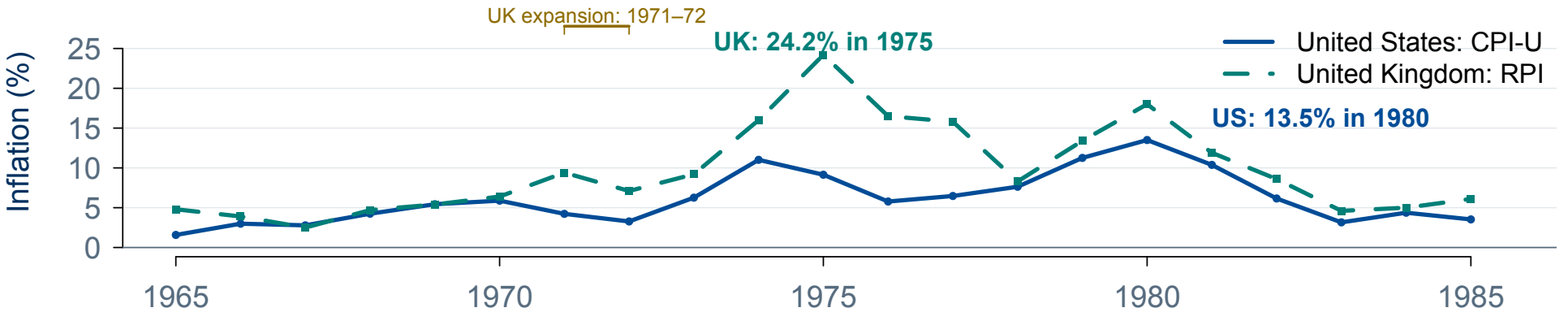
Oil costs raise inflation at each unemployment rate. Falling costs reverse the shift.

B: inflation and unemployment rise

Firms reduce output and hiring. Workers may seek wage increases to restore lost purchasing power.

Federal Reserve Board, Waller (2023), The Unstable Phillips Curve · RBA, Causes of Inflation

Oil Shocks and Stagflation



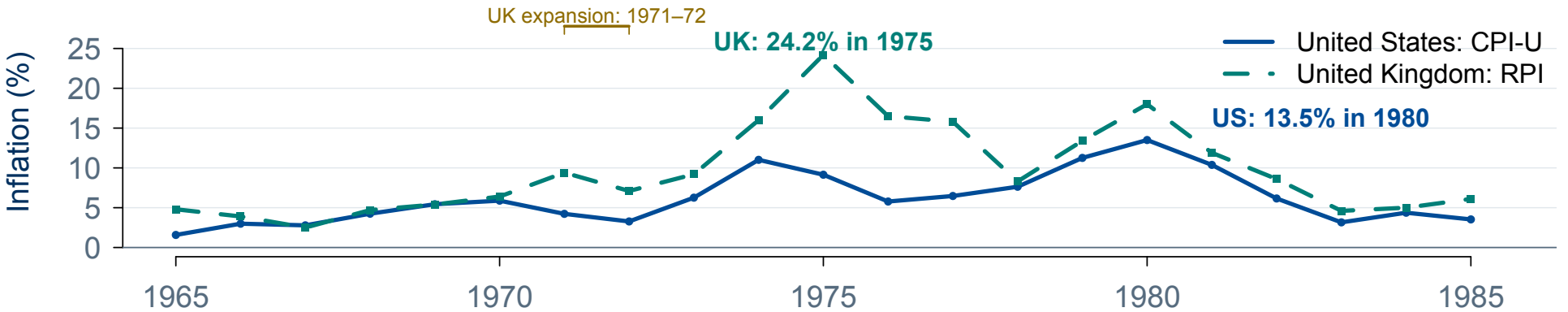
Annual averages. Source: CBO February 2026 historical CPI-U file (BLS); ONS CZBH, RPI. Different national indexes.

Contains ONS information, Crown copyright, licensed under the Open Government Licence v3.0.

1973–1975: after the 1973–74 oil embargo

US inflation: 6.3% → 9.1%; unemployment: 4.9% → 8.5%.

Oil Shocks and Stagflation



Annual averages. Source: CBO February 2026 historical CPI-U file (BLS); ONS CZBH, RPI. Different national indexes.

Contains ONS information, Crown copyright, licensed under the Open Government Licence v3.0.

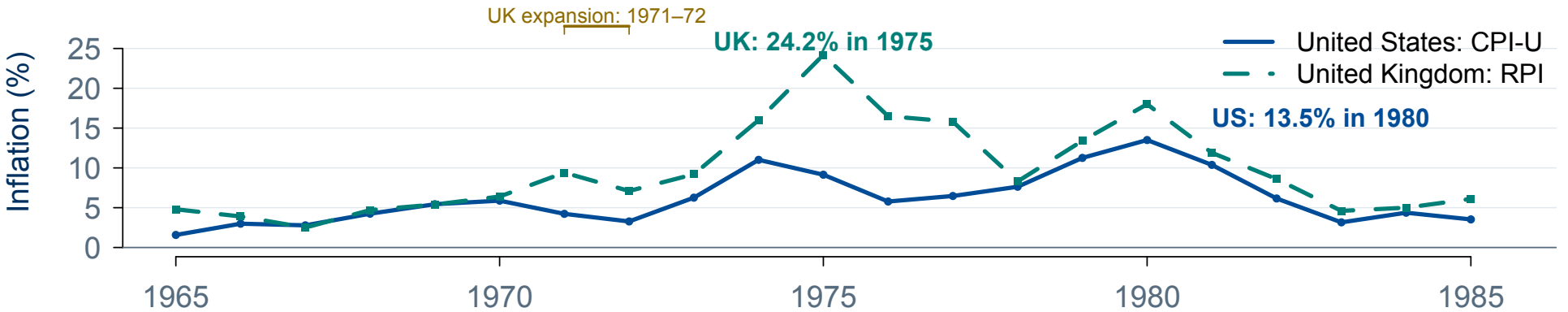
1973–1975: after the 1973–74 oil embargo

US inflation: 6.3% → 9.1%; unemployment: 4.9% → 8.5%.

1978–1980: the Iranian revolution and second oil shock

US inflation: 7.6% → 13.5%; unemployment: 6.1% → 7.2%.

Oil Shocks and Stagflation



Annual averages. Source: CBO February 2026 historical CPI-U file (BLS); ONS CZBH, RPI. Different national indexes.

Contains ONS information, Crown copyright, licensed under the Open Government Licence v3.0.

1973–1975: after the 1973–74 oil embargo

US inflation: 6.3% → 9.1%; unemployment: 4.9% → 8.5%.

1978–1980: the Iranian revolution and second oil shock

US inflation: 7.6% → 13.5%; unemployment: 6.1% → 7.2%.

Stagflation

A supply shift can raise inflation and unemployment together while output weakens.

Expected Inflation

Definition

The inflation rate that employers and workers expect in the near future.

Lansing, FRBSF (2002), Can the Phillips Curve Help Forecast Inflation? · Federal Reserve Board, Waller (2023), The Unstable Phillips Curve

Expected Inflation

Definition

The inflation rate that employers and workers expect in the near future.

Friedman and Phelps

Their analyses in the late 1960s made inflation expectations central to the unemployment–inflation relationship.

Lansing, FRBSF (2002), Can the Phillips Curve Help Forecast Inflation? · Federal Reserve Board, Waller (2023), The Unstable Phillips Curve

Expected Inflation

Definition

The inflation rate that employers and workers expect in the near future.

Friedman and Phelps

Their analyses in the late 1960s made inflation expectations central to the unemployment–inflation relationship.

Future expectations, present decisions

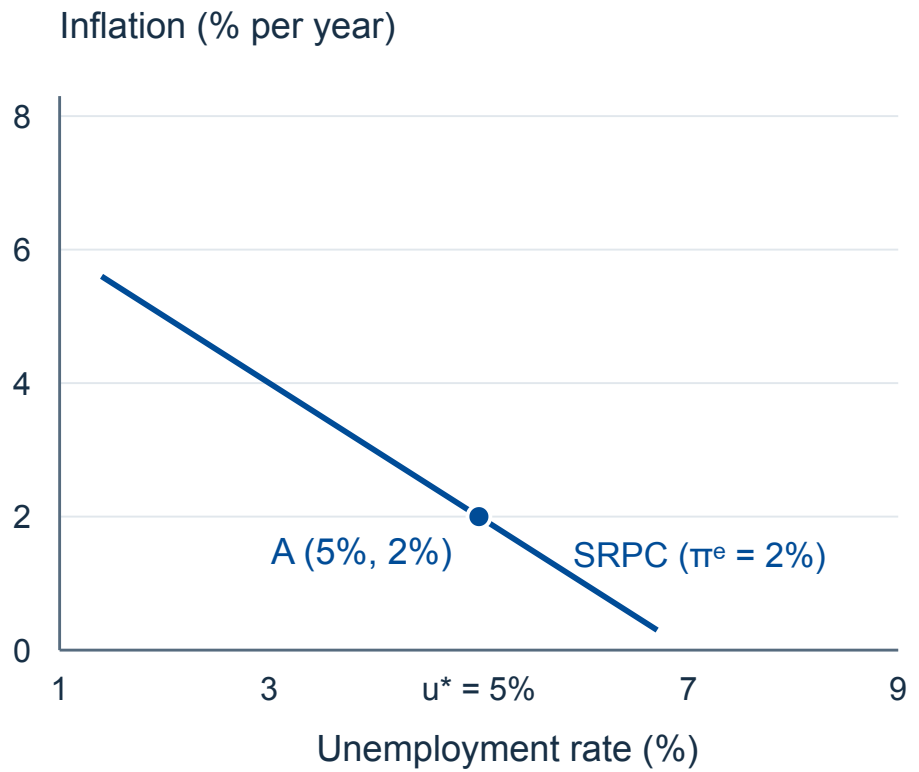
Expected price increases affect wage bargaining and price setting today, influencing current inflation.

Lansing, FRBSF (2002), Can the Phillips Curve Help Forecast Inflation? · Federal Reserve Board, Waller (2023), The Unstable Phillips Curve

16.3 Inflation and Unemployment in the Long Run

- Explain why expectations adjustment leads to a vertical long-run Phillips curve at NAIRU.

Inflation Expectations and the Short-Run Phillips Curve

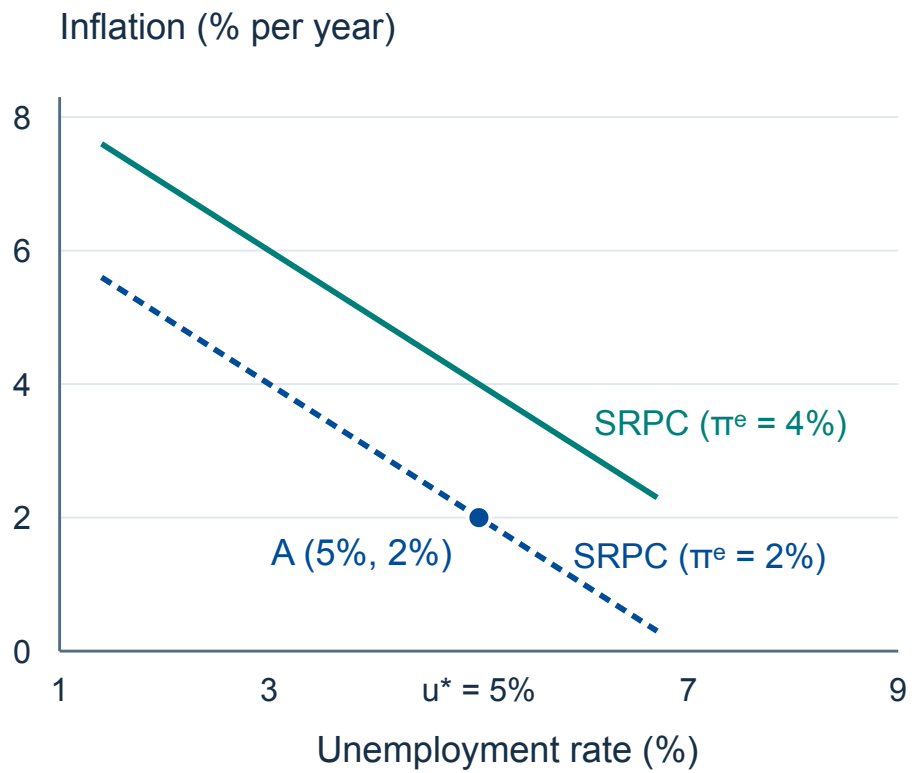


A: expected inflation of 2%

At 5% unemployment, actual and expected inflation are both 2%.

Federal Reserve Board, Waller (2023), The Unstable Phillips Curve · Lansing, FRBSF (2002), Can the Phillips Curve Help Forecast Inflation?

Inflation Expectations and the Short-Run Phillips Curve



A: expected inflation of 2%

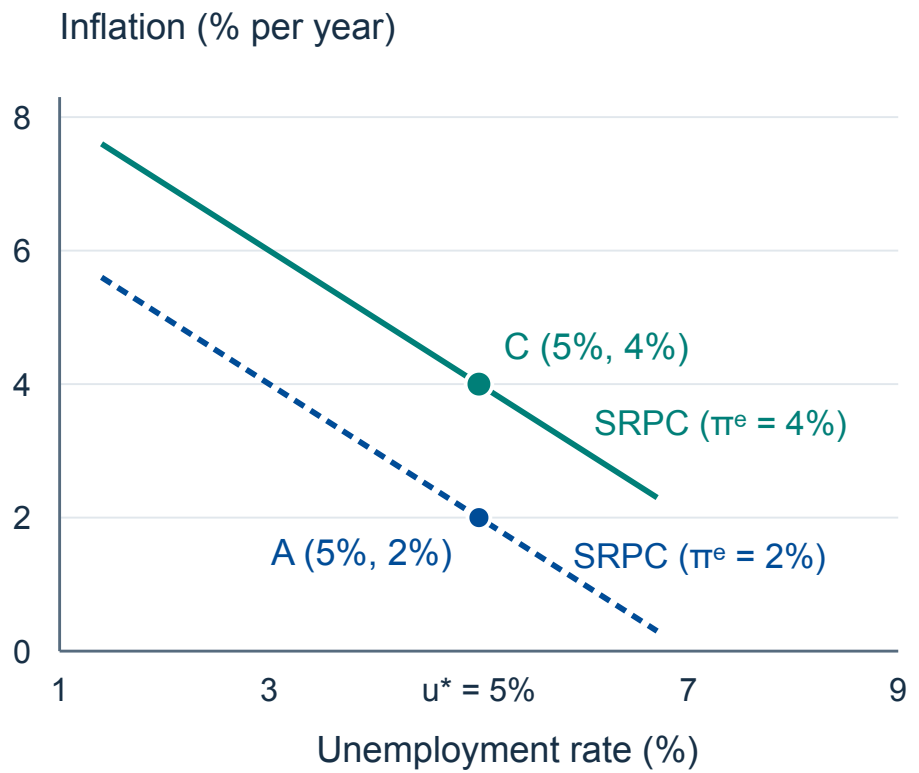
At 5% unemployment, actual and expected inflation are both 2%.

Persistent inflation changes expectations

As 4% inflation becomes expected, wage negotiations and price decisions incorporate it.

Federal Reserve Board, Waller (2023), The Unstable Phillips Curve · Lansing, FRBSF (2002), Can the Phillips Curve Help Forecast Inflation?

Inflation Expectations and the Short-Run Phillips Curve



A: expected inflation of 2%

At 5% unemployment, actual and expected inflation are both 2%.

Persistent inflation changes expectations

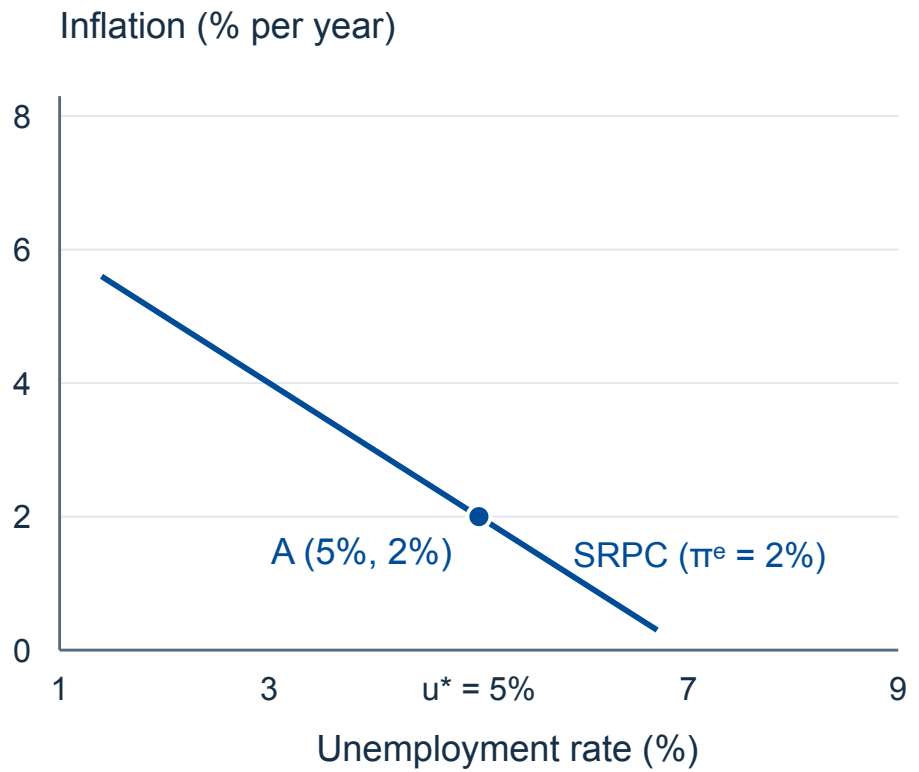
As 4% inflation becomes expected, wage negotiations and price decisions incorporate it.

An upward shift

The new short-run curve has higher inflation at each unemployment rate; at 5%, inflation is now 4%.

Federal Reserve Board, Waller (2023), The Unstable Phillips Curve · Lansing, FRBSF (2002), Can the Phillips Curve Help Forecast Inflation?

Adjustment to the Long Run

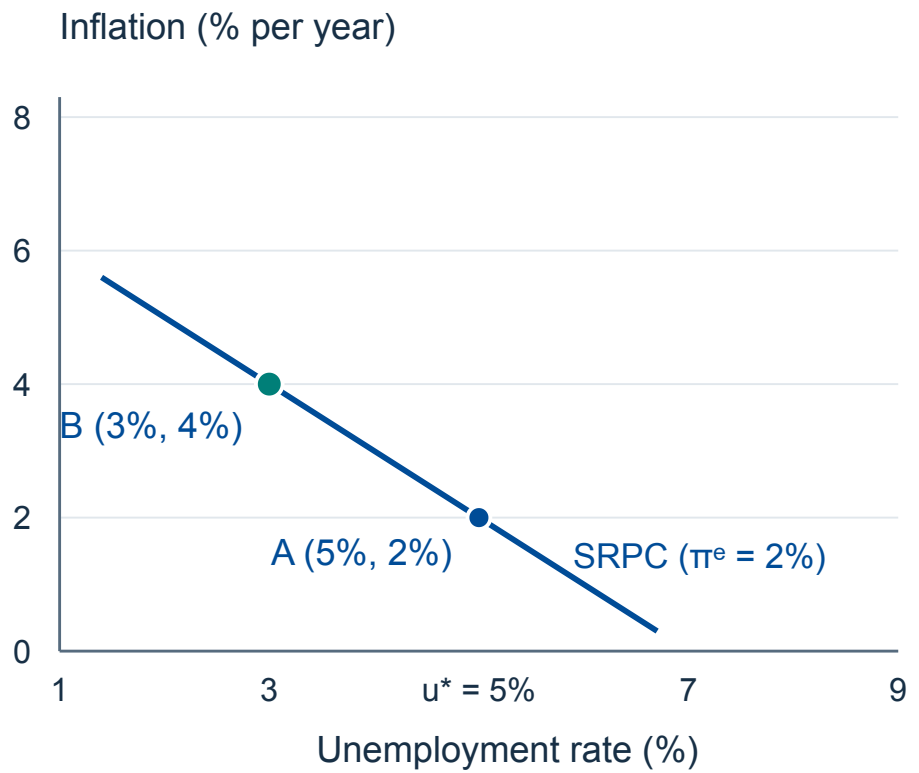


A: initial long-run equilibrium

Unemployment is 5%; actual and expected inflation are 2%.

Federal Reserve Board, Waller (2023), The Unstable Phillips Curve · Federal Reserve Board, Mishkin (2007), Estimating Potential Output

Adjustment to the Long Run



A: initial long-run equilibrium

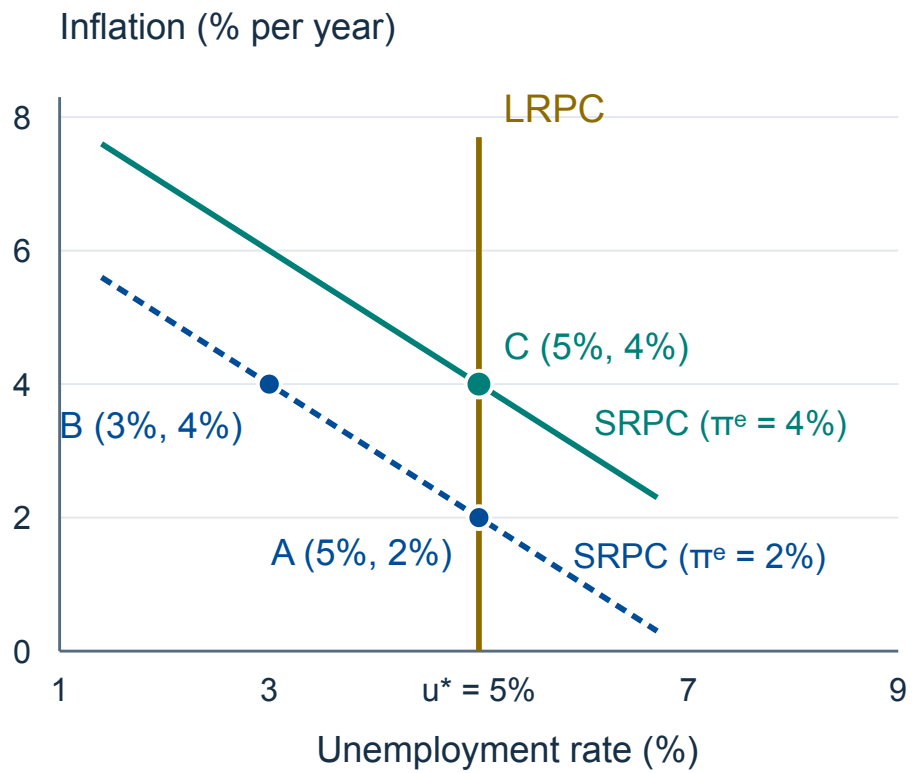
Unemployment is 5%; actual and expected inflation are 2%.

B: short-run demand expansion

Inflation rises to 4% before expectations adjust. Unemployment temporarily falls to 3%.

Federal Reserve Board, Waller (2023), The Unstable Phillips Curve · Federal Reserve Board, Mishkin (2007), Estimating Potential Output

Adjustment to the Long Run



A: initial long-run equilibrium

Unemployment is 5%; actual and expected inflation are 2%.

B: short-run demand expansion

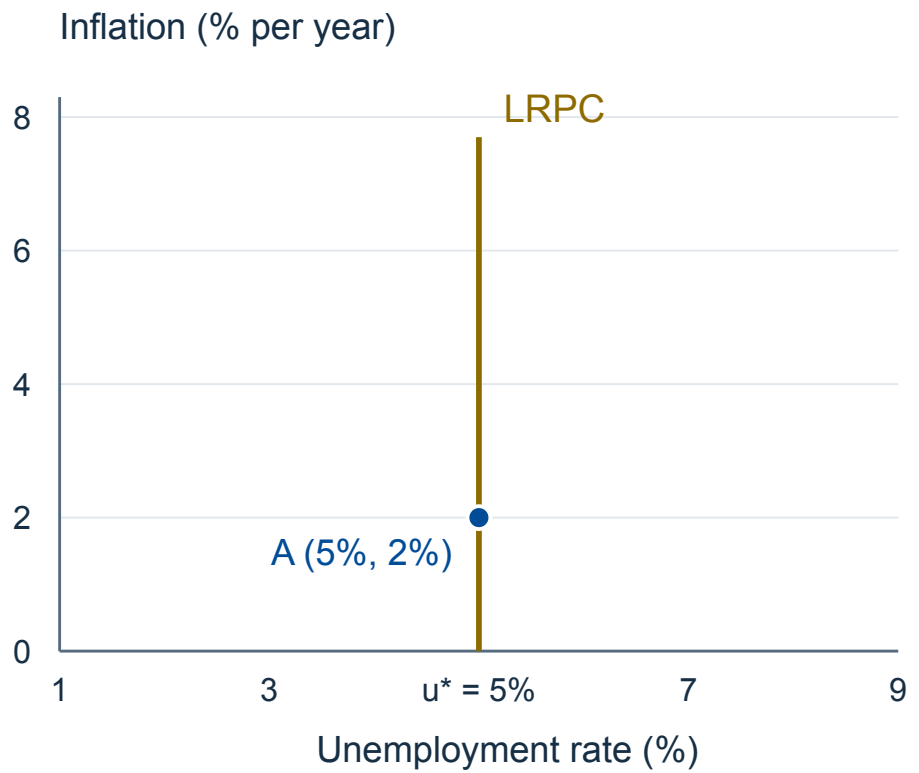
Inflation rises to 4% before expectations adjust. Unemployment temporarily falls to 3%.

C: expectations catch up

Wages and prices incorporate 4% inflation. The short-run curve shifts up; unemployment returns to 5%.

Federal Reserve Board, Waller (2023), The Unstable Phillips Curve · Federal Reserve Board, Mishkin (2007), Estimating Potential Output

The Long-Run Phillips Curve and NAIRU

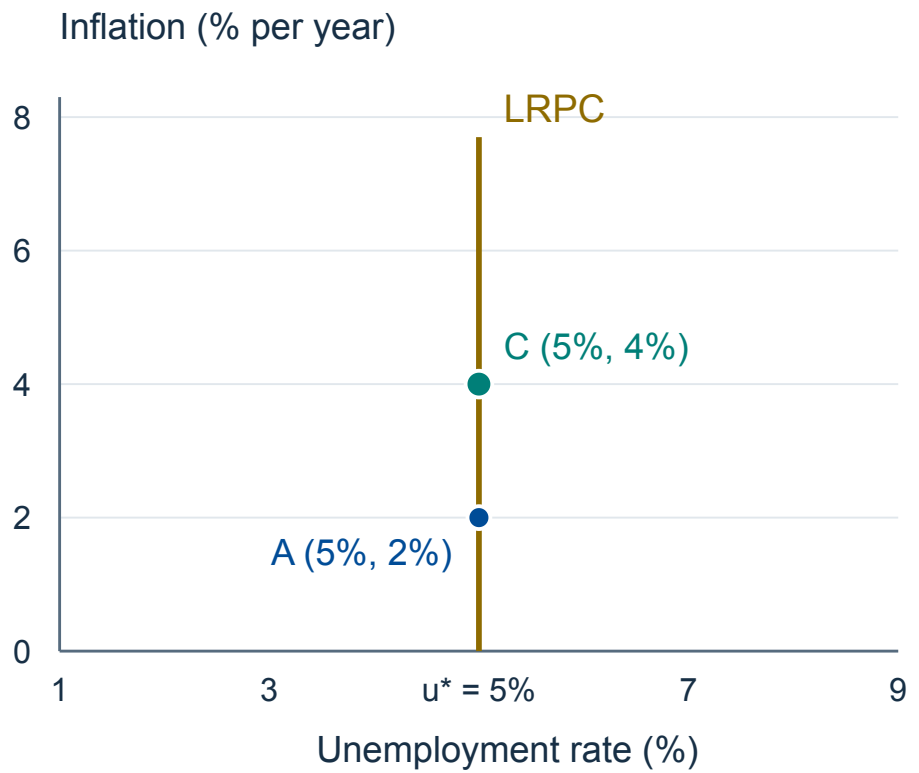


A vertical long-run curve

Stable inflation of 2% or 4% has the same u* = 5%. Higher fully expected inflation cannot permanently lower unemployment.

RBA, The Non-Accelerating Inflation Rate of Unemployment · Federal Reserve Board, Mishkin (2007), Estimating Potential Output · Federal Reserve Board, Waller (2023), The Unstable Phillips Curve

The Long-Run Phillips Curve and NAIRU



A vertical long-run curve

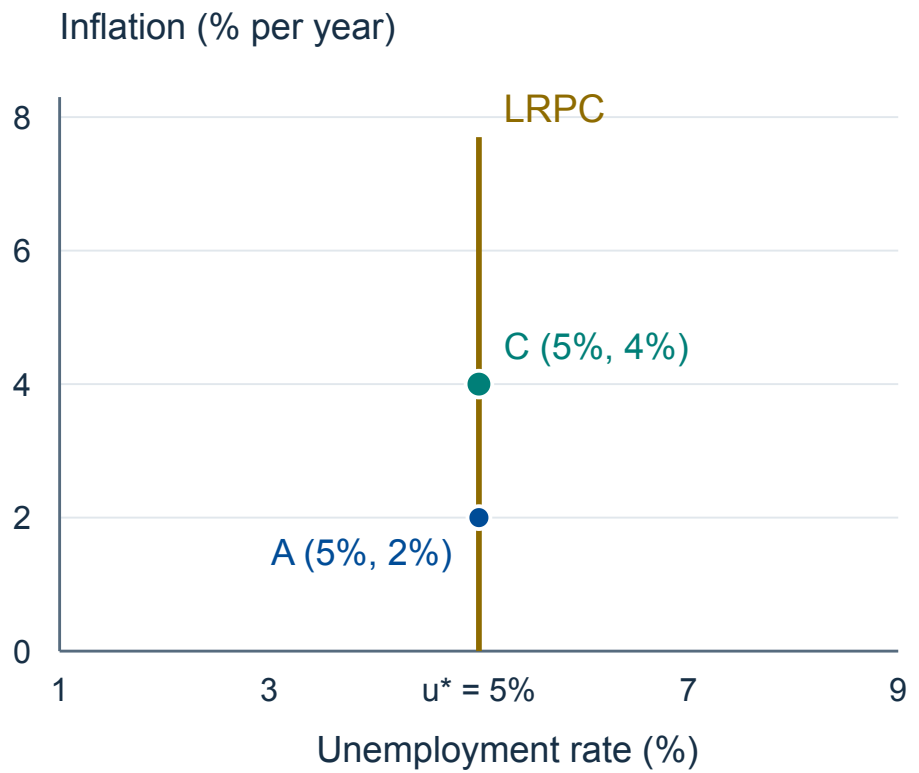
Stable inflation of 2% or 4% has the same u* = 5%. Higher fully expected inflation cannot permanently lower unemployment.

NAIRU

Non-Accelerating Inflation Rate of Unemployment: the natural rate in this model, consistent with stable inflation, other conditions fixed.

RBA, The Non-Accelerating Inflation Rate of Unemployment · Federal Reserve Board, Mishkin (2007), Estimating Potential Output · Federal Reserve Board, Waller (2023), The Unstable Phillips Curve

The Long-Run Phillips Curve and NAIRU



A vertical long-run curve

Stable inflation of 2% or 4% has the same u* = 5%. Higher fully expected inflation cannot permanently lower unemployment.

NAIRU

Non-Accelerating Inflation Rate of Unemployment: the natural rate in this model, consistent with stable inflation, other conditions fixed.

Inflation pressure

Below u*, inflation tends to rise; above it, to fall. At u*, cyclical unemployment is zero; frictional and structural unemployment remain.

RBA, The Non-Accelerating Inflation Rate of Unemployment · Federal Reserve Board, Mishkin (2007), Estimating Potential Output · Federal Reserve Board, Waller (2023), The Unstable Phillips Curve